

Week – 1

Q. A local library is currently managing its books and members using a physical notebook, which has resulted in several operational problems such as misplaced books, difficulty tracking borrowed items, and inaccurate member records. To modernize its operations, the library plans to develop a database system called BookFlow. As a database developer, your task is to create the SQL foundation for this system. First, create a new database named `bookflow_db`. Then, design two tables: a `Books` table containing the columns `book_id`, `title`, `isbn`, and `published_year`, and a `Members` table containing the columns `member_id`, `full_name`, and `email`. Ensure data integrity by setting `book_id` and `member_id` as PRIMARY KEYS, applying a NOT NULL constraint to the `title` column so that every book must have a name, enforcing UNIQUE constraints on both `isbn` and `email` to prevent duplicate entries, and adding a CHECK constraint to ensure that the `published_year` is not in the future (for example, it must be less than 2027). Finally, insert three sample records into both the `Books` and `Members` tables to verify that the database structure and constraints are functioning correctly. This exercise will help you develop skills in schema definition, data integrity enforcement, and the use of SQL Data Definition Language (DDL) commands such as `CREATE DATABASE`, `CREATE TABLE`, and `INSERT` statements.

Execution :

```
Enter password: ****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 20
Server version: 8.0.46 MySQL Community Server - GPL

Copyright (c) 2000, 2026, Oracle and/or its affiliates.

Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE bookflow_db;
Query OK, 1 row affected (0.01 sec)

mysql> USE bookflow_db;
Database changed
mysql> CREATE TABLE books (
  -> book_id INT AUTO_INCREMENT PRIMARY KEY,
  -> title VARCHAR(255) NOT NULL,
  -> isbn VARCHAR(13) NOT NULL UNIQUE,
  -> published_year INT,
  -> CONSTRAINT chk_published_year CHECK (published_year < 2027));
Query OK, 0 rows affected (0.02 sec)
```

```
Enter password: ****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 20
Server version: 8.0.46 MySQL Community Server - GPL

Copyright (c) 2000, 2026, Oracle and/or its affiliates.

Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE bookflow_db;
Query OK, 1 row affected (0.01 sec)

mysql> USE bookflow_db;
Database changed
mysql> CREATE TABLE books (
  -> book_id INT AUTO_INCREMENT PRIMARY KEY,
  -> title VARCHAR(255) NOT NULL,
  -> isbn VARCHAR(13) NOT NULL UNIQUE,
  -> published_year INT,
  -> CONSTRAINT chk_published_year CHECK (published_year < 2027));
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> CREATE TABLE members (  
-> member_id INT AUTO_INCREMENT PRIMARY KEY,  
-> full_name VARCHAR(100) NOT NULL,  
-> email VARCHAR(150) NOT NULL UNIQUE);  
Query OK, 0 rows affected (0.01 sec)  
  
mysql> INSERT INTO books (title, isbn, published_year) VALUES  
-> ('The Alchemist', '9780061122415', 1988),  
-> ('Clean Code', '9780132350884', 2008),  
-> ('Atomic Habits', '9780735211292', 2018)  
-> ;  
Query OK, 3 rows affected (0.00 sec)  
Records: 3 Duplicates: 0 Warnings: 0  
  
mysql> INSERT INTO members (full_name, email) VALUES  
-> ('Anil Kumar', 'anil.kumar@example.com'),  
-> ('Priya Sharma', 'priya.sharma@example.com'),  
-> ('Ravi Verma', 'ravi.verma@example.com');  
Query OK, 3 rows affected (0.01 sec)  
Records: 3 Duplicates: 0 Warnings: 0
```